

SECTION 2 – PREINSTALLATION

2–1 ENVIRONMENTAL REQUIREMENTS

STORAGE PRIOR TO INSTALLATION. Store Lightning PDU only in an environment which meets the following conditions:

- Temperature: –30 to 140 °F
–17 to 60 °C
- Humidity: 20% to 80% non–condensing
- Altitude: 0 to 8000 ft (0 to 2440 m)

OPERATING ENVIRONMENT. Provide an operating environment which meets the following conditions:

- Ambient temperature: 50 to 80 °F
10 to 27 °C
- Ambient temperature change: 5 °F (2.8 °C) per hour
- Ambient temperature gradient: maximum 5 °F (2.8 °C)
- Relative humidity: 30% to 60% non–condensing,
maximum rate of change of 5% per hour
- Magnetic field: maximum 200 gauss
- Altitude: 0 to 8000 ft (0 to 2440 m)

2–2 FLOOR LOADING

Lightning PDU has a base of 22.1 X 30 in. (56 X 76 cm). Provide a floor with strength sufficient for loads specified in Table 2–1.

TABLE 2–1
FLOOR LOADING

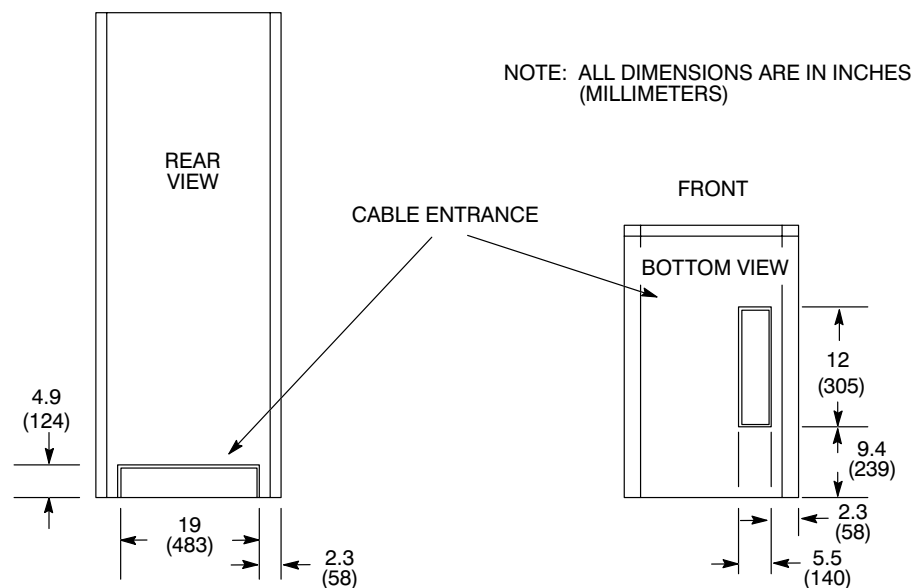
MODEL	FLOOR LOADING
46–317099P20 and P21	233 lbs/sq. ft. 1,143 kg/sq. m.

2-3 FLOOR LAYOUT AND CABLE ACCESS

Locate Lightning PDU close to power source to minimize line inductance.

Interconnection cables may be routed into Lightning PDU through rear panel, or on right side of bottom panel. See Illustration 2-1.

After installation, minimum service access area for Lightning PDU is 3 ft (91 cm) in the front. Some service is easier with access to either side panel, but side access is not required. See Illustration 2-2.



CABLE ENTRANCE
ILLUSTRATION 2-1

2-4 SEISMIC INFORMATION

Lightning PDU center of gravity is shown in Illustration 2-2. See Table 2-2 for dimensions.

For a mobile installation, or if required by local seismic code, mount Lightning PDU to floor using two angle brackets. Mount top of Lightning PDU to wall on either side or back. Use a wall angle bracket.

2-4 SEISMIC INFORMATION (continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES (MILLIMETERS)
- APPROX. WEIGHT: 1000 lbs. (455 kg)
-  INDICATES AIR FLOW
-  INDICATES CENTER OF GRAVITY. DIMENSIONS A, B, C SEE TABLE 2-2.

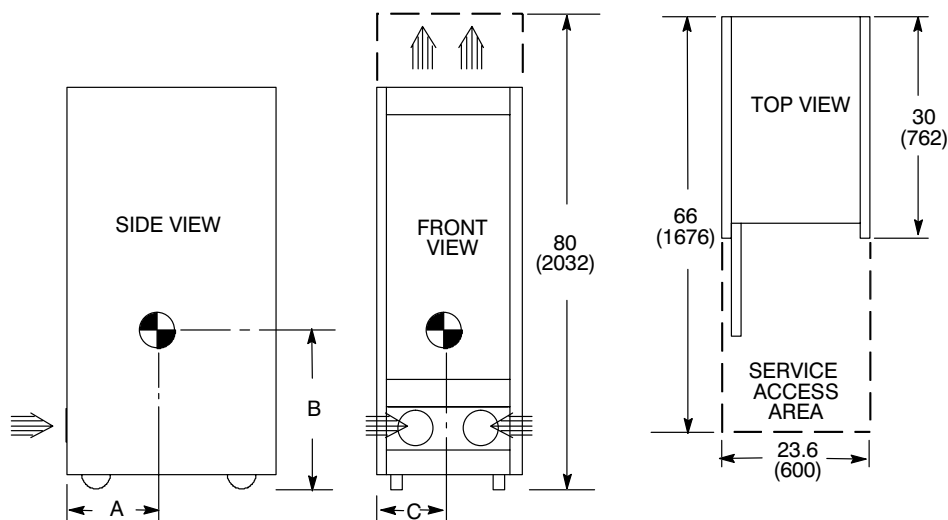
**SERVICE AND VENTILATION CLEARANCE**

ILLUSTRATION 2-2

TABLE 2-2
CENTER OF GRAVITY

DIMENSION (SEE ILLUS. 2-2)	MODEL
	DIMENSIONS ARE IN INCHES (MILLIMETERS)
	46-317099P20 AND P21
A	17.4 (442)
B	18.7 (475)
C	12.5 (318)

2-5 ROUTE TO INSTALLATION SITE

Lightning PDU weighs about 1000 lbs (455 kg). Use care when rolling Lightning PDU. It weighs 250 lbs (114 kg) per caster. Plywood or steel plates may be used to distribute weight and protect floor surface from overload damage.

2–5 ROUTE TO INSTALLATION SITE (continued)

Lightning PDU can be rolled on integral casters to the installation site with protective cardboard packaging still in place, all doors and corridors must be 24 X 71 in. (61 X 181 cm) or larger. For Lightning PDU dimensions refer to Table 2–3.

Any elevator en route to installation site must be of a size and weight–bearing capacity to carry Lightning PDU. With protective cardboard packaging in place, each Lightning PDU with voltage regulation 970 lbs (440 kg). For net weight refer to Table 2–3.

TABLE 2–3
SITE PLANNING SPECIFICATIONS

MODEL 46–317099 >	P20	P21
NOMINAL INPUT VOLTAGE	480	380/40 0/415
INPUT VOLTAGE RANGE	418–504	331–399 348–420 361–436
MAXIMUM INPUT VOLTAGE	504	436
NOMINAL RATED CURRENT (A)	66	80
MAXIMUM RATED CURRENT (A)	80	
FREQUENCY (Hz)	60	50/60
COOLING REQUIREMENT (BTU/hr)	7000	
AMBIENT TEMPERATURE F (C)	80 (26.7)	
THE FOLLOWING ARE WITHOUT PACKAGING:		
SIZE H X W X D in. (cm)	70 X 22.1 X 30 (1779 X 56 X 76)	
WEIGHT lbs (kg)	854 (388)	933 (424)

2–6 DELIVERY INFORMATION

All Lightning PDU's are wrapped in plastic and shipped on integral casters.

Roll to installation site on integral casters.